

Senyi Kong

Atmospheric optics | Satellite remote sensing | Data assimilation

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ResearchGate | Expected PhD completion: June 2027

PROFILE

PhD researcher seeking research and postdoctoral opportunities in atmospheric science. Experience connects particle-scale scattering physics with satellite retrieval and numerical weather prediction, including all-sky infrared assimilation and Bayesian-machine learning methods.

EDUCATION

PhD in Earth Climate and Environment | Zhejiang University

Sep 2022 - Jun 2027 (expected) | Direct-entry doctoral programme. All doctoral coursework completed.

BSc in Atmospheric Science | Zhejiang University

Sep 2018 - Jun 2022 | GPA 4.60; ranked first in class.

SELECTED RESEARCH EXPERIENCE

All-sky hyperspectral data assimilation

FY-4B GIIRS assimilation in CMA-GFS 4DVar; detector and cloud-bias treatment, observation-error modelling, and forecast evaluation. First author of the two-part 2026 QJRM study.

Physics-informed atmospheric retrieval

Hybrid Bayesian-machine learning retrieval of atmospheric and cloud profiles from GIIRS observations. First author, *Advances in Atmospheric Sciences*, 2026.

Electromagnetic scattering and particle optics

Nonspherical dust and cloud-particle modelling, IITM scattering calculations, and optical-property uncertainty analysis; coauthor of the ZJU-SCATT database.

SELECTED FIRST-AUTHOR PUBLICATIONS

1. **S. Kong**, W. Han, L. Bi, R. Yin (2026). All-sky FY4B-GIIRS assimilation in CMA-GFS 4DVar, Part I: Implementation. *QJRM*. DOI
2. **S. Kong**, W. Han, L. Bi, R. Yin (2026). All-sky FY4B-GIIRS assimilation in CMA-GFS 4DVar, Part II: Evaluation. *QJRM*. DOI
3. **S. Kong**, L. Bi, W. Han, R. Yin, H. Zhang (2026). Hybrid Bayesian-machine learning retrieval of atmospheric profiles. *Advances in Atmospheric Sciences*. DOI
4. **S. Kong**, Z. Wang, L. Bi (2024). Uncertainty in dust refractive-index retrieval and optical properties. *Atmospheric Chemistry and Physics*. DOI

Titles abbreviated; DOI links lead to the full publications. Complete bibliography in the academic CV.

SELECTED RECOGNITION AND PATENT

National Scholarship; Zhejiang University First-Class Scholarship; Outstanding Graduate, Zhejiang Province; Top Ten Doctoral Research Advances (2023); Top Ten Poster Award (2022).

Patent: Second-listed inventor, iterative detector non-uniformity bias correction. Application CN202511848925.6; filed December 2025; granted.

METHODS AND LANGUAGE

IITM and electromagnetic scattering; radiative transfer; hyperspectral retrieval; 4DVar assimilation; Bayesian and machine-learning methods. English: CET-6, 592.

Collaboration: Research-group peer support and volunteer contributions to student activities.